www.tntextbooks.in

There are different types of equilibrium.

Physical equilibrium

Chemical Equilibrium

- Chemical reactions which are reversible do not cease, when equilibrium is attained. At equilibrium the forward and the backward reactions are proceeding at the same rate and no macroscopic change is observed. So chemical equilibrium is in a state of dynamic equilibrium.

- The rate law states that, "At any instant, the rate of a chemical reaction at a given temperature is directly proportional to the product of the active masses of the reactants at that instant".

- The value of equilibrium constant, Kc tells us the extent of a reaction, i.e., it indicates how far the reaction has proceeded towards product formation at a given temperature.

- Under non-equilibrium conditions, reaction quotient 'Q' is defined as the ratio of the product of active masses of reaction products raised to the respective stoichiometric coefficients in the balanced chemical equation to that of the reactants.

- Le-Chatelier's Principle "If a system at equilibrium is disturbed, then the system shifts itself in a direction that nullifies the effect of that disturbance."

- The effect of change in reaction conditions such as temperature, pressure and concentration etc… on a system at equilibrium can be predicted by Le Chatelier-Braun principle.

- vant Hoff equation This equation gives the quantitative temperature dependence of equilibrium constant (K).

Evaluation

1. If K_b and K_f for a reversible reactions are 0.8×10^{-5} and 1.6×10^{-4} respectively, the value of the equilibrium constant is,

   a) 20 b) 0.2×10^{-1}

   c) 0.05 d) none of these

2. At a given temperature and pressure, the equilibrium constant values for the equilibria

   $3A_2 + B_2 + 2C \overset{K_1}{\rightleftharpoons} 2A_3BC$ and

   $A_3BC \overset{K_2}{\rightleftharpoons} \frac{3}{2}[A_2] + \frac{1}{2}B_2 + C$

   The relation between K_1 and K_2 is

   a) $K_1 = \frac{1}{\sqrt{K_2}}$ b) $K_2 = K_1^{-1/2}$

   c) $K_1^2 = 2K_2$ d) $\frac{K_1}{2} = K_2$

3. The equilibrium constant for a reaction at room temperature is K1 and that at 700 K is K2. If K1 > K2, then

   a) The forward reaction is exothermic